

Solving Equations

1 The Apollonius Problem

- circles touching three given circles
- systems of polynomial equations

2 Groebner Bases

- a lexicographic term order triangulates

MCS 320 Lecture 29
Introduction to Symbolic Computation
Jan Verschelde, 22 July 2026

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1 The Apollonius Problem

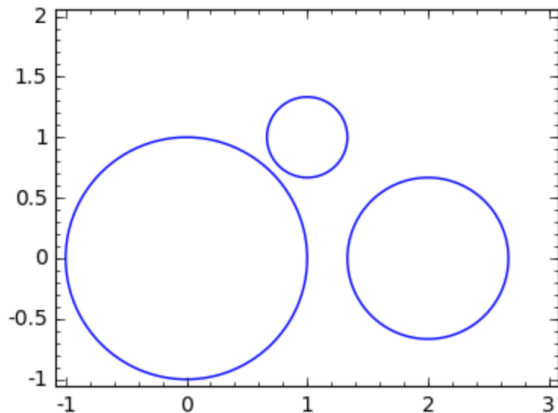
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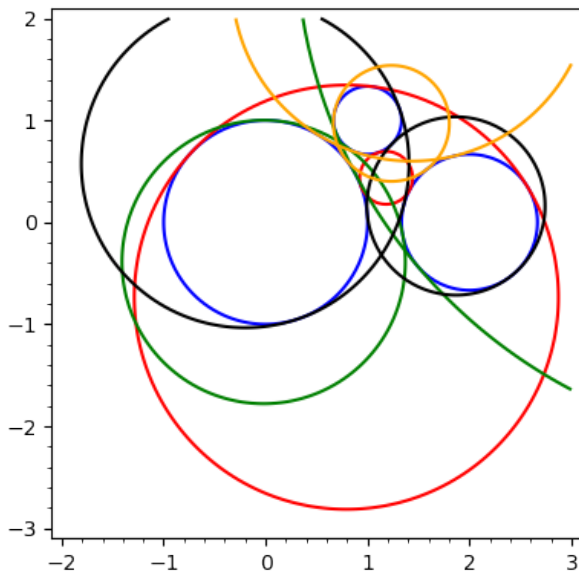
The Apollonius Problem

Given are three circles:



Compute all circles that touch these three circles.

All Circles touching the Three Given Circles



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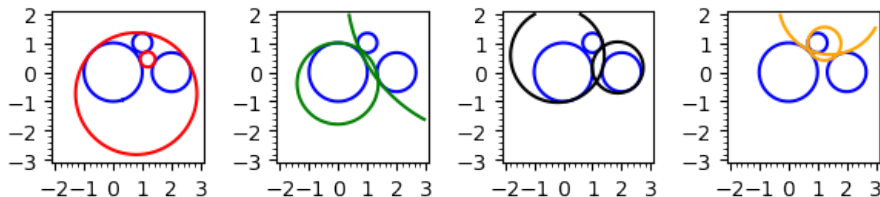
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All Circles touching the Three Given Circles

The touching circles are computed in pairs:



The center (x, y) and radius r of the first pair are the solutions of

$$\begin{cases} (x - c_{1,x})^2 + (y - c_{1,y})^2 - (r - r_1)^2 = 0 \\ (x - c_{2,x})^2 + (y - c_{2,y})^2 - (r - r_2)^2 = 0 \\ (x - c_{3,x})^2 + (y - c_{3,y})^2 - (r - r_3)^2 = 0 \end{cases}$$

where the three circles are defined by the coordinates of the centers $(c_{1,x}, c_{1,y})$, $(c_{2,x}, c_{2,y})$, $(c_{3,x}, c_{3,y})$ with corresponding radii r_1 , r_2 , and r_3 .

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a Lexicographic Term Order Triangulates

The polynomials in the system are ordered lexicographically.

Given a system, a Groebner basis generalizes

- the Euclidean algorithm for computing greatest common divisors of polynomials in one variable, and
- Gaussian elimination for linear systems.

For the problem of Apollonius, the three input circles are defined by the centers $(0, 0)$, $(2, 0)$, $(1, 1)$ and radii 1, $2/3$, and $1/3$.

The Groebner basis for the first pair of solutions is then

$$x + \frac{1}{6}r - \frac{41}{36}, \quad y + \frac{1}{2}r - \frac{11}{36}, \quad r^2 - \frac{71}{39}r - \frac{253}{468},$$

which shows there are indeed two solutions.